DunenScope

User manual

Version 0.7.0 — November 2026

DunenScope is a Windows tool for diagnosing and commissioning Dunen / ELYX motor controllers. It uses the same protocol as DunenConfiger and complements it.

1. Overview

DunenScope is used to read, understand and fine-tune a Dunen / ELYX controller from a Windows PC, during machine commissioning or after-sales service. It is intended for machine builders who integrate these controllers and for their technicians.

DunenConfiger remains the reference configuration tool. DunenScope reads the same parameter file, with the same row numbers, and its xlsx exports can be re-imported into DunenConfiger. It adds monitoring, diagnostic and tuning functions:

Area	What DunenScope adds
Plots	Six synchronised plots, a custom plot with up to 4 variables of your choice, CSV recording
Dashboard	Speed, torque, voltage, current, throttle, power, temperatures, state and faults on one screen
Diagnostics	"Why is it capped?": which limit is active (torque, battery current, speed, voltage) and for how long; battery, wired inputs and temperatures; last fault analysis
Setup wizard	Step-by-step assistant: motor, controller and battery, command and pedal calibration, machine, writing, encoder zero, direction of rotation, PI tuning
Simple settings	The whole controller by category, in the machine's units (s, %, km/h, L/min, m/min), mouse-editable curves, assistants
Comparison	Differences against a parameter file; end-of-line test with report and production log
Support	Complete support package in one click (.zip)
Speed PI tuning	Measurement assistant on the machine, manual step tuning and automatic tuning
Profiles	Application profiles (machine + CAN adapter) that can be exported to another PC
Languages	Interface in 10 languages: French, English, Chinese, Spanish, German, Italian, Portuguese (Brazil), Russian, Japanese, Korean
CAN adapters	PEAK PCAN-USB, Kvaser, IXXAT, Vector, SLCAN (CANable...)

2. Installation

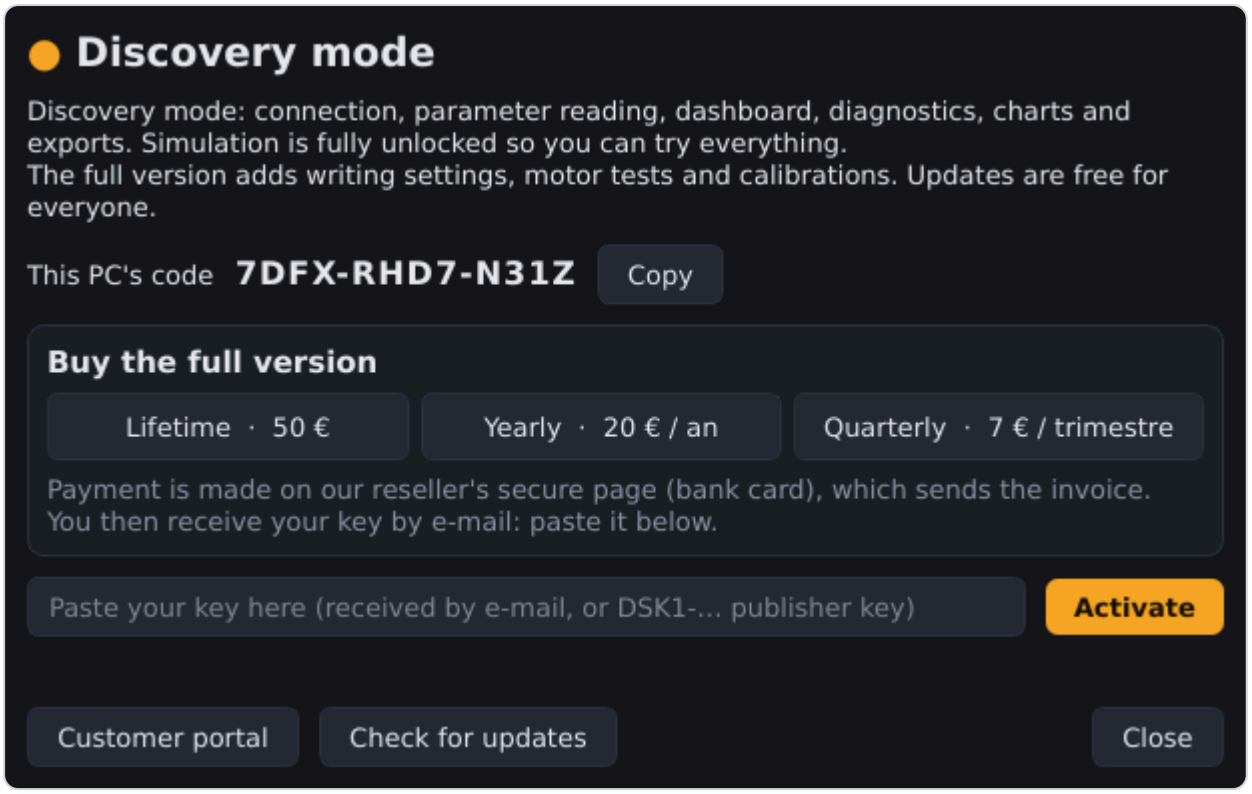
- A single file to run: **DunenScope_Setup_0.7.0.exe**. A setup wizard in 10 languages (language, licence agreement, folder, shortcuts) creates the shortcut and the manual in the Start menu; it can be uninstalled from Windows "Apps". The application is delivered compiled and ready to use.
- A new version installs over the old one, manually or automatically (section 3): settings, profiles and backups are kept.
- The interface is available in 10 languages (list at the top right): French, English, Chinese, Spanish, German, Italian, Portuguese (Brazil), Russian, Japanese, Korean. By default, DunenScope uses the Windows display language (first entry of the list, "(auto)"); choosing a specific language keeps it from then on.
- If the PEAK driver for the PCAN-USB adapter is not found, the installer offers to install it. DunenScope also offers it at connection time if it is missing.
- For the other adapters (Kvaser, IXXAT, Vector, SLCAN), install the driver supplied by their manufacturer.
- System requirements: Windows 10 or Windows 11, 64-bit.

3. Licence, purchase and updates

DunenScope installs and runs free of charge in **discovery mode**: connection, parameter reading, dashboard, diagnostics, charts, exports and support package. In **simulation**, every feature is unlocked so you can try it. The **full version** adds everything that writes to a real controller (settings, motor tests, calibrations). Updates are free for everyone, with or without a key.

The button at the top right ("Licence: discovery" or "Licence: full") opens the Licence window:

Item	Purpose
This PC's code	Short PC identifier (12 characters), to give when asking for a key tied to this PC
Buy	Lifetime, Yearly or Quarterly: opens the secure payment page; the key arrives by e-mail
Activate	Paste the key received (purchased key, or DSK1-... publisher key), then "Activate"
Release this PC	Makes the key usable on another PC (internet required)
Customer portal	Invoices, subscription, activated PCs
Check for updates	Checks right away whether a new version has been published



Licence window (discovery mode)

- **One key = one PC.** A purchased key is activated once over the internet (a phone's hotspot is enough) and stays tied to that PC. On another PC it is refused until the first one has been released.
- **Without internet.** Lifetime purchase: DunenScope then works without internet. Subscription: an online check is needed at least every 45 days; DunenScope warns 10 days ahead.
- **Publisher keys (DSK1-...).** Supplied by the publisher (trials, partners): checked without internet, optionally limited to one PC and/or until a date.
- **End of licence.** Subscription stopped, key disabled or expired: DunenScope returns to discovery mode; backups, profiles and files are kept.
- **Automatic updates (free for everyone).** With internet, in discovery mode as well as in the full version, DunenScope checks at start-up whether a new version has been published and offers it ("Install now", "Later", "Skip this version"). The installer is downloaded and its signature checked (an altered file is refused), then it replaces the old version and restarts DunenScope. Windows asks for permission to install.

4. First connection

1. Key ON, adapter plugged into the controller's CAN connector.
2. Leave the top bar on **"Adapter: auto"** and **"Controller: auto"**, then click **"Connect"** (or the same button in the dashboard banner; "Try without hardware (simulation)" starts the simulation). DunenScope finds by itself the adapter that is plugged in (PEAK PCAN-USB, Kvaser, IXXAT, Vector or CANable), the bus bit rate and the controller address (node). The search only reads: nothing is written to the controller.

3. The indicator next to the button changes to "Connected". The status bar at the bottom shows "Link OK" and the number of samples per second (in orange if the link becomes unstable), then the model, software and hardware versions and the serial number; the log shows what was found (for example "PCAN_USBBUS1, 250 kbit/s, node 1").
4. The model is recognised and its parameter file loaded automatically. The model can also be chosen in the "Controller" list.

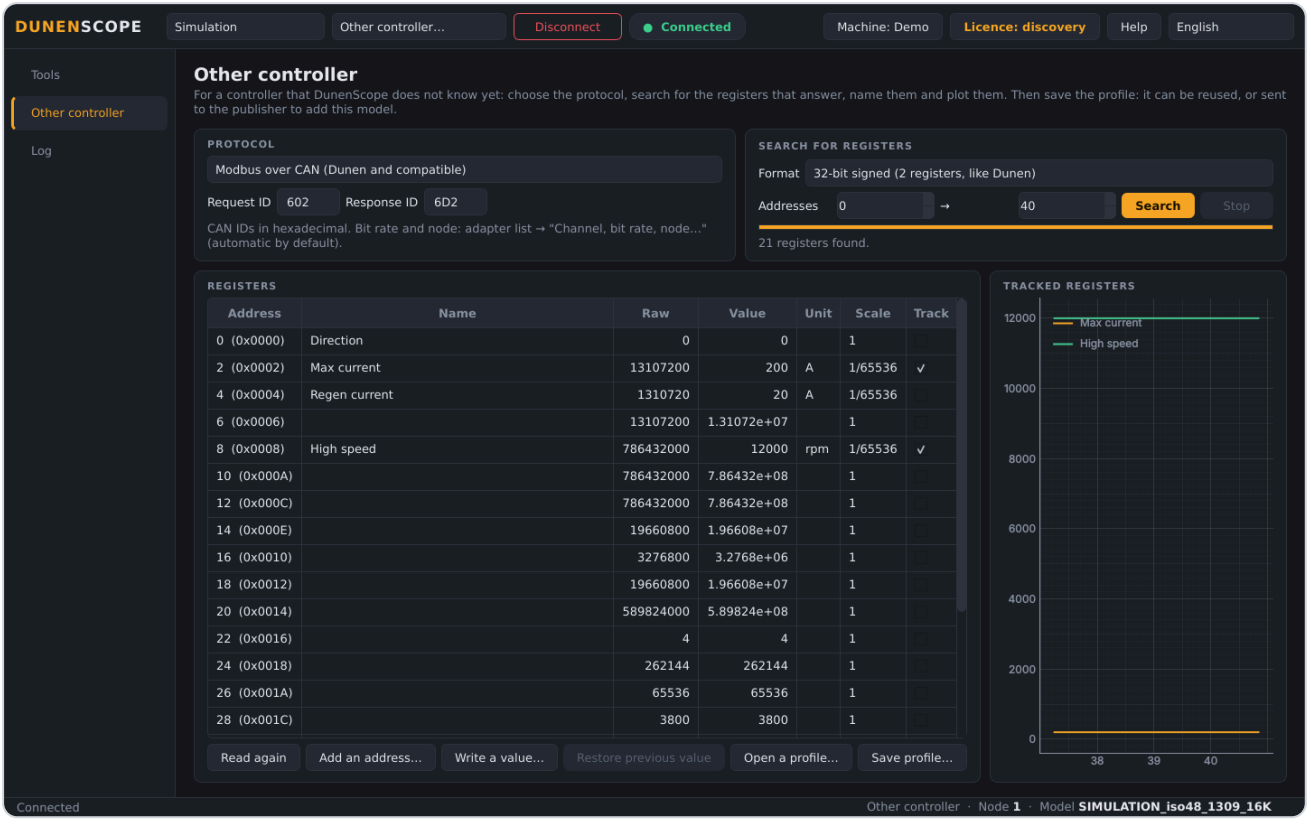
The last choice of the adapter list, "Channel, bit rate, node...", sets the channel, bit rate and node by hand, which is only useful when several controllers share the same bus. The next connection first tries the settings that worked last time.

DunenScope and DunenConfiger cannot use the adapter at the same time: close one before opening the other.

To try the application without hardware, select the "Simulation (no hardware)" adapter: a simulated controller feeds every page. The screenshots in this manual were taken in this mode.

Unknown controller: "Other controller" mode

If the connected controller is not yet known to DunenScope, it stays **read-only** so that nothing is damaged, and offers the "Other controller" mode. This mode can also be chosen in the "Controller" list.



Other controller

1. Choose the protocol (today: Modbus over CAN, like Dunen controllers) and, if needed, the CAN request and response identifiers.
2. Choose the value format (16 or 32 bits, signed or not) and the address range, then "Search": DunenScope reads the registers that answer, without writing anything.

3. Give the registers found a name, a unit and a scale (fixed-point values: type 1/65536 for the Q16 format used by Dunen controllers, 1/10 for tenths); tick "Track" to see them on a chart.
4. "Write a value..." (full version): confirmation, old value kept, read-back; "Restore previous value" undoes it.
5. "Save profile..." keeps this work for next time. Sending the profile and the support package to the publisher lets this controller be added to DunenScope.

Help, contact and error reports

The **Help** button in the top bar opens the "Help and contact" window: version, manual, software page and contact address. DunenScope is brand new: to report a problem, suggest an idea or ask a question, write directly in this window and click "Send". The technical information and the session log are attached (untick the box if needed, "Show what will be sent" shows everything); no personal data is sent. Without internet, the message is kept and sent at the next start.

If an unexpected error occurs, or if DunenScope closed abruptly, a window offers to send the error report. Nothing is sent without your consent; the "Always send error reports without asking" box skips the question next time.

5. Dashboard

The Dashboard continuously shows the main values read from the controller.



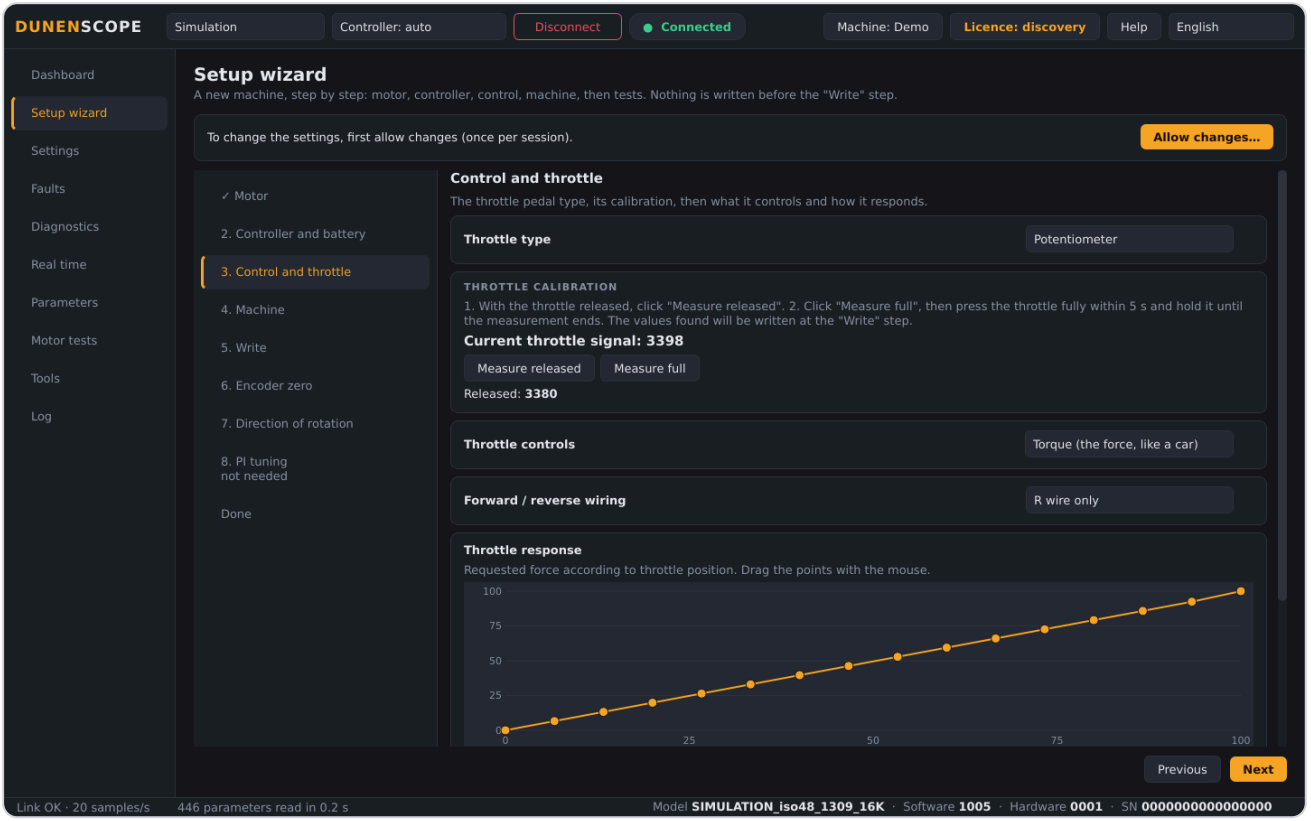
Dashboard

- **Four gauges**: motor speed with its setpoint, motor torque with its limit, battery voltage with the power-reduction and undervoltage cut-off thresholds, battery current with its limit. The scales follow the motor's capabilities.
- **Throttle** in %, with the raw signal (ADC) and the configured released and full-travel thresholds.
- **Estimated mechanical power**, and electrical power drawn from the battery.
- **Application tile**, depending on the machine type: hydraulic pump (speed and theoretical flow, as on the screenshot), vehicle speed in km/h, cable speed and pull of a winch, conveyor speed, fan or compressor speed after the gearbox, or electrical power.
- **State** (Running, direction, time powered), current **fault / warning**, motor and controller **temperatures** with their power-reduction threshold.
- **Current controller settings**: control mode, maximum speed, maximum torque, maximum battery current, undervoltage cut-off; "All settings →" opens the Settings page.
- **Banner** (while nothing is connected): reminds what to do, with "Connect" and "Try without hardware (simulation)".

The status bar shows "Link OK · n samples/s", or "Unstable link" in orange when requests go unanswered; the details (requests, errors, time per request) appear on hover.

6. Setup wizard

The **Setup wizard** tab guides the configuration of a new machine, step by step, like the VESC wizard. "Next" and "Previous" move between steps; a finished step is ticked ✓. Nothing is written before the "Write" step.



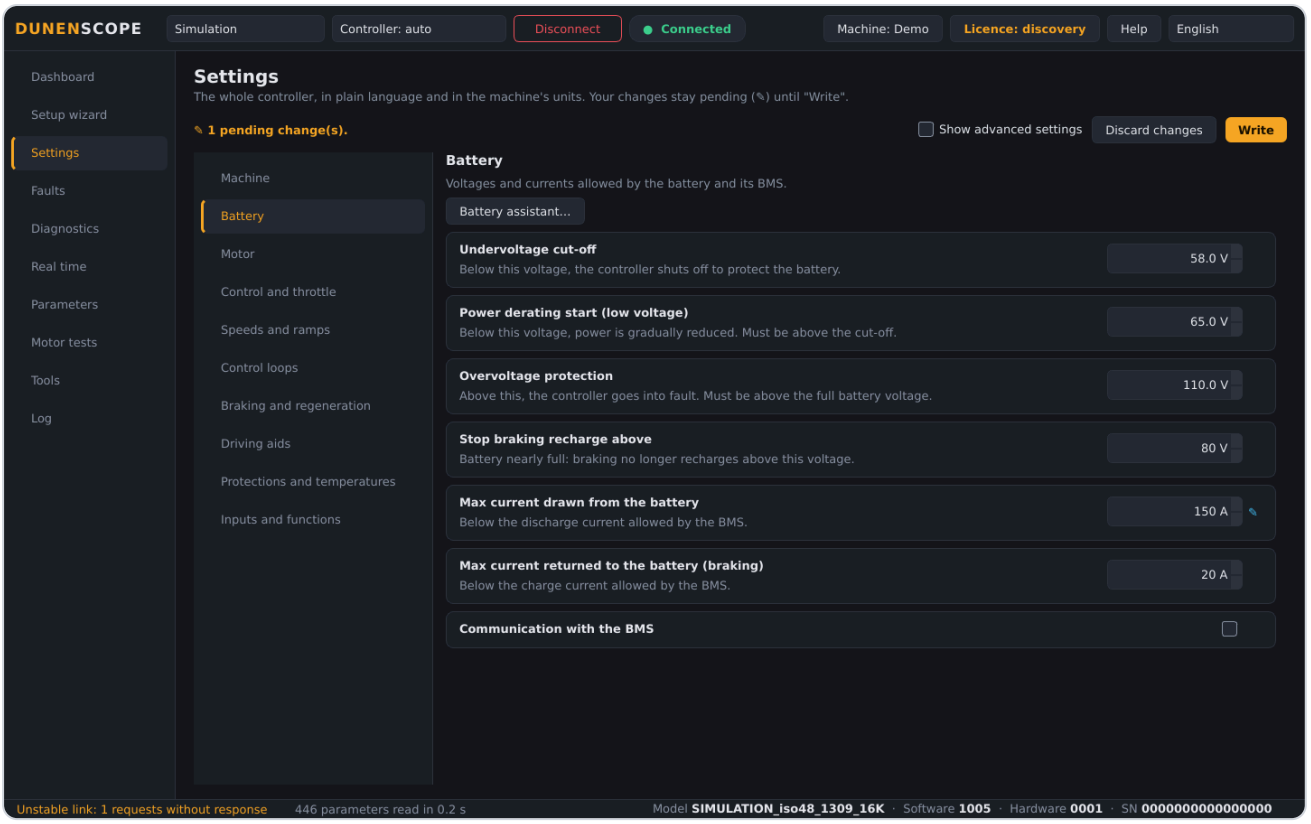
Setup wizard

1. **Motor**: pole pairs, maximum torque, maximum phase current, temperature sensor and derating threshold, position sensor (type, pulses per revolution, direction).
2. **Controller and battery**: detected controller (model, versions), voltage thresholds, currents allowed by the battery and its BMS; "Battery assistant..." calculates them from the battery type and number of cells.
3. **Control and throttle**: pedal type, on-the-spot calibration ("Measure released", then "Measure full": a 5 s countdown leaves time to press the pedal fully, then the measurement runs; the signal is shown live), what the pedal controls (torque or speed), forward / reverse wiring, pedal response and torque ramp smoothness.
4. **Machine**: machine type, reduction ratio, wheel, drum or pump (kept on this PC), then maximum speed in the machine's units (with its equivalent in motor rpm) and, when the pedal controls speed, the times to reach it and to come back to a stop.
5. **Write**: list of pending changes in plain words, then "Write to the controller" (full backup before, read-back after).
6. **Encoder zero**: motor uncoupled from its load, the controller turns it slowly to find the zero of its position sensor; DunenScope says whether the zero was found, then "Keep this zero in the controller" saves it. If you move on without keeping it, DunenScope offers to.
7. **Direction of rotation**: a slow 5 s forward rotation; "Yes, right direction" or "No, reverse the direction" (the direction is then reversed and written: just repeat the test). The question is asked even if you stopped the test with STOP, as long as the motor turned; if it did not turn, DunenScope says so.
8. **PI tuning**: only when the pedal controls speed (step greyed out otherwise). With the motor mounted on the machine with its real load, "Auto-tune the PI..." opens the Motor tests tab; otherwise, skip the step and tune the PI later.
9. **Done**: summary (✓ done, ○ not done, or not needed) and final check: "Disconnect", key off for 10 s, key on, then "Connect". DunenScope reads the settings again and compares them with what was written: "✓ n setting(s) written, kept", or the list of those lost, prepared again, with a button to go back to the "Write" step. Connecting another controller resets the wizard.

Safety: the motor tests (steps 6 and 7) only start after the "Machine safe" box is ticked and once all changes are written; the STOP button and the Esc key cut the motor immediately, and each test stops by itself. In discovery mode, the wizard can be browsed and reads the values; changing them requires the full version ("View offers..." banner). Otherwise, the "Allow changes..." banner allows them, with confirmation.

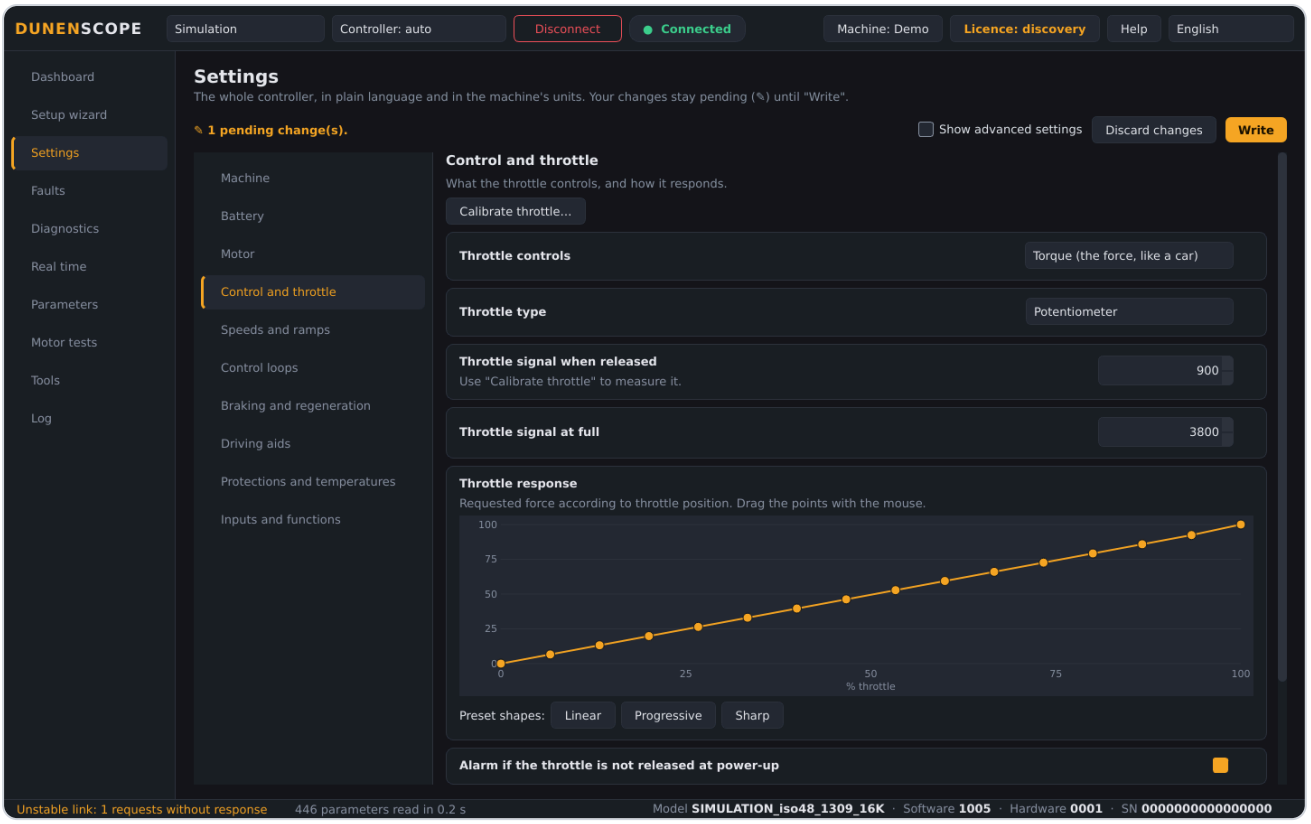
7. Settings

The **Settings** page shows the whole controller sorted by category, in plain language and in the machine's units. It is the simplest way to tune a machine: no need to know the parameter names.



Settings

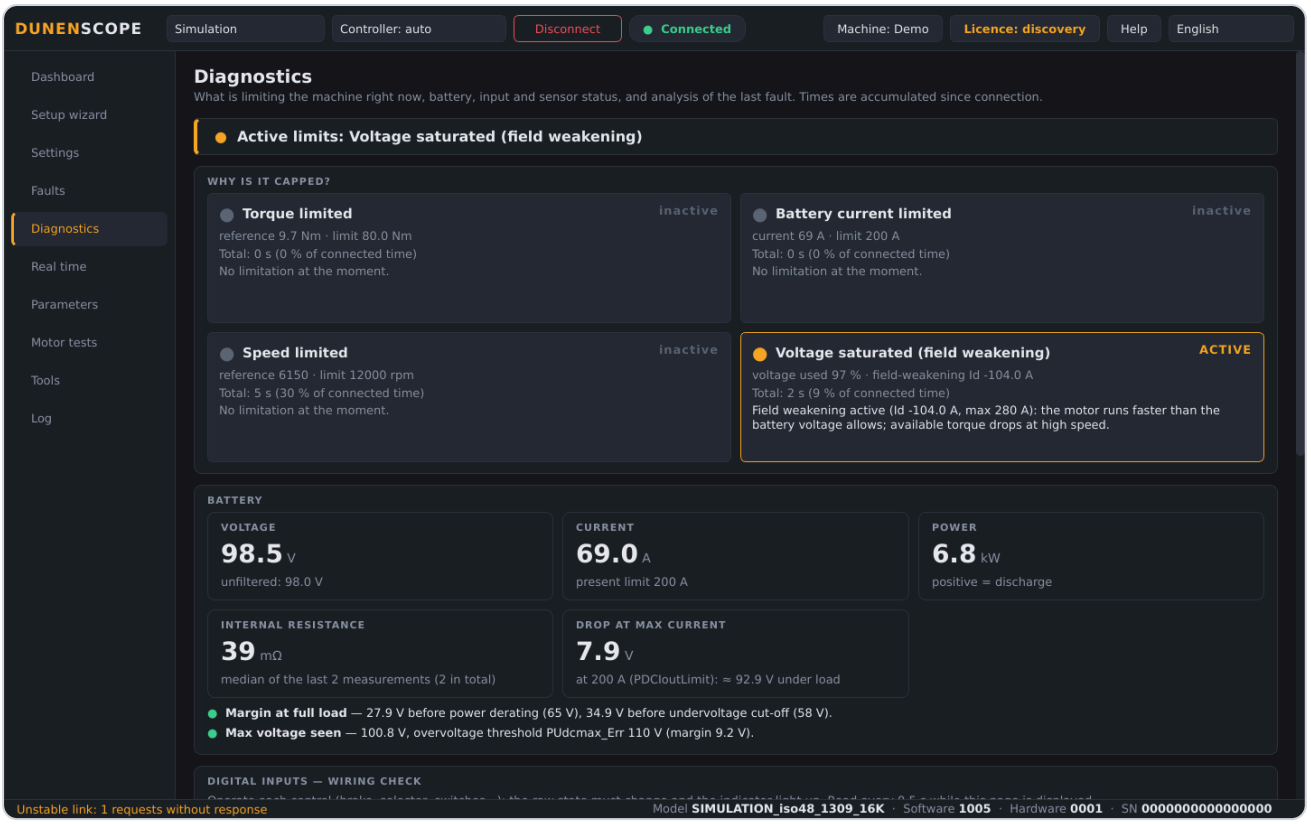
- **Categories**: Machine, Battery, Motor, Control and throttle, Speeds and ramps, Control loops, Braking and regeneration, Driving aids, Protections and temperatures, Inputs and functions.
- **Reading**: when the page opens, DunenScope reads the controller settings if this has not been done yet.
- **Machine units**: ramps in seconds (time to reach maximum torque or speed), coefficients in %, speeds in km/h (vehicle), L/min (pump), m/min (winch, conveyor) or rpm, depending on the machine type chosen in the Machine category.
- **Only what matters**: settings that are useless in the chosen mode are hidden (for example speed ramps when the throttle controls torque), as are those the connected model does not have. "Show advanced settings" shows the rest.
- **Curves**: throttle response and braking curves are changed by dragging the points with the mouse; "Linear", "Progressive" and "Sharp" give ready-made shapes.
- **Assistants**: "Battery assistant..." (easy setup), "Calibrate throttle..." (opens the "Control and throttle" step of the Setup wizard), "Encoder auto-learning...", "Auto-tune the PI...".
- **Nothing is written straight away**: every change is prepared in the parameter list (✎ rows). The first change asks you to allow changes (confirmation; full version for a real controller). "Write" goes through the usual steps: consistency check, full backup, writing, saving and read-back. "Discard changes" clears what was prepared.



Throttle response

8. Diagnostics

The Diagnostics page shows what is limiting the machine right now. A banner lists the active limits; on the screenshot, "No active limit".



Diagnostics

- **Why is it capped?** Four cards in a 2×2 grid: "Torque limited", "Battery current limited", "Speed limited", "Voltage saturated (field weakening)". Each shows the reference and the limit, the time accumulated since connection and the cause.
- **Battery**: voltage, current, power, estimated internal resistance, expected voltage drop at maximum current, margin before power reduction and undervoltage cut-off, maximum voltage seen against the overvoltage threshold.
- **Digital inputs — wiring check**: configured function, active level and raw state of each input. Operate each control (brake, selector, switches…): the raw state must change.
- **Thermal**: motor and controller temperatures, raw motor sensor value, estimated time left before the power-reduction threshold.
- **Last fault analysis**: reads the conditions recorded at the last fault and gives the probable cause.

9. Real time

The Real time page plots the main quantities on six synchronised plots over an adjustable window (30 s on the screenshot). Mouse wheel = zoom, right-click = options.

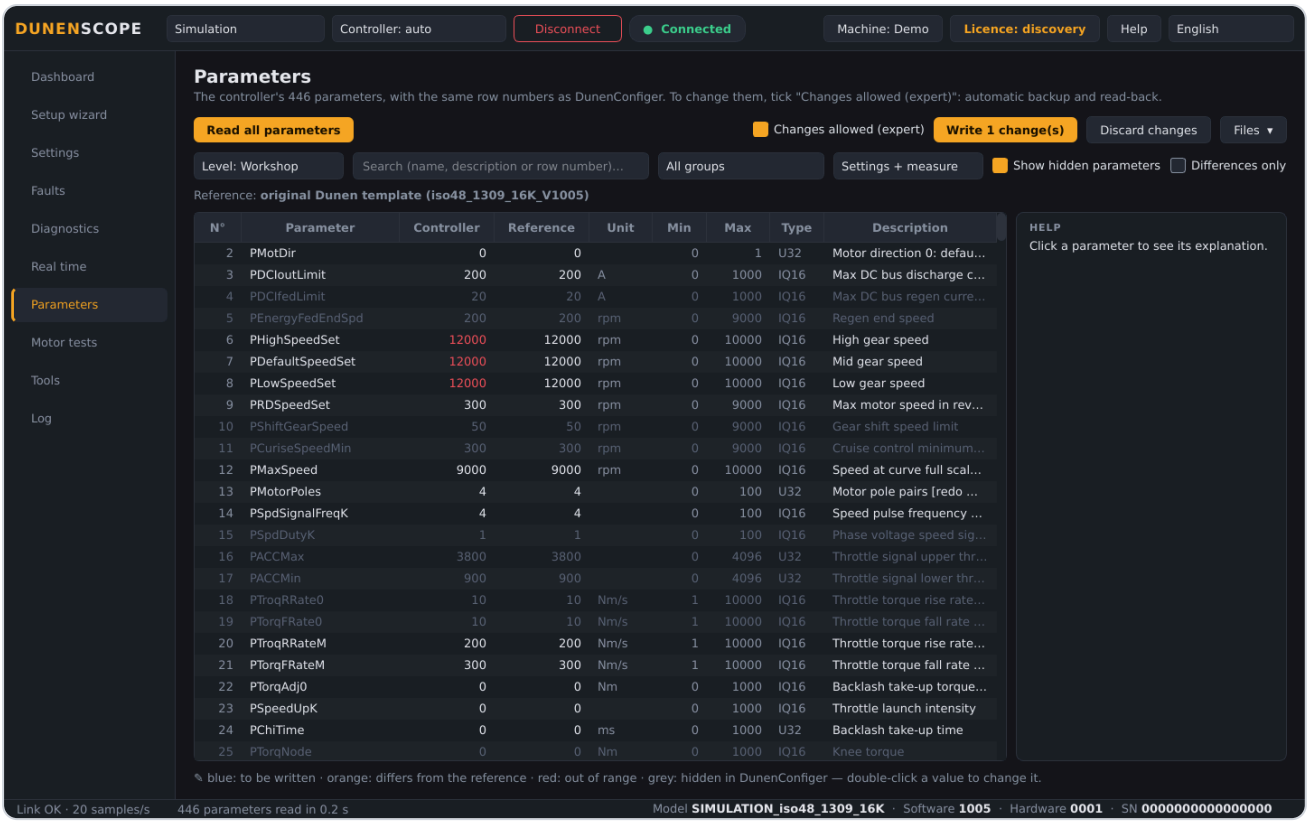


Real time

- **Plots**: measured speed and setpoint; torque and limit; battery current and phase current; battery voltage; throttle pedal and temperatures; custom plot.
- **Custom plot**: "+ Add a variable to the custom plot...", up to 4 variables of your choice (IADin2 on the screenshot). Double-clicking a row on the Parameters page also adds it.
- **Record CSV**: records the measurements to a file during the test. "Pause" freezes the display, "Clear" empties it.
- **Speed PI tuning assistant**: click "Measure the next setpoint change", press the pedal decisively (for example from half to full) and hold it for 3 seconds. DunenScope compares the measured speed with the setpoint and shows the lag on the ramp, overshoot, settling time, final error, ripple and number of oscillations, followed by a recommendation. On the screenshot: final error +71 rpm, recommendation "steady-state error: raise Ki by 20–50 %". This assistant writes nothing to the controller.

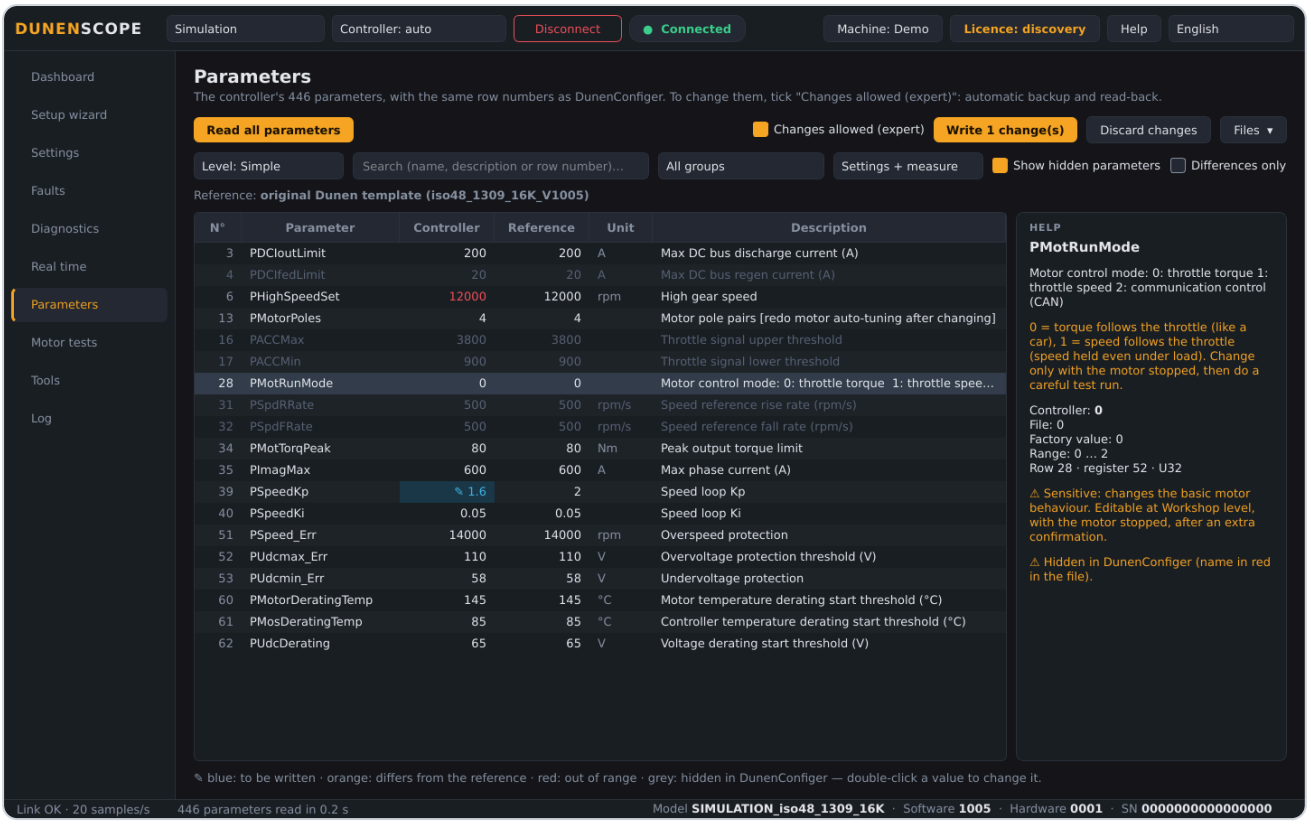
10. Parameters

The Parameters page lists the controller's 446 parameters, with the same row numbers as DunenConfiger. The screenshot shows changes allowed, with one pending change.



Parameters

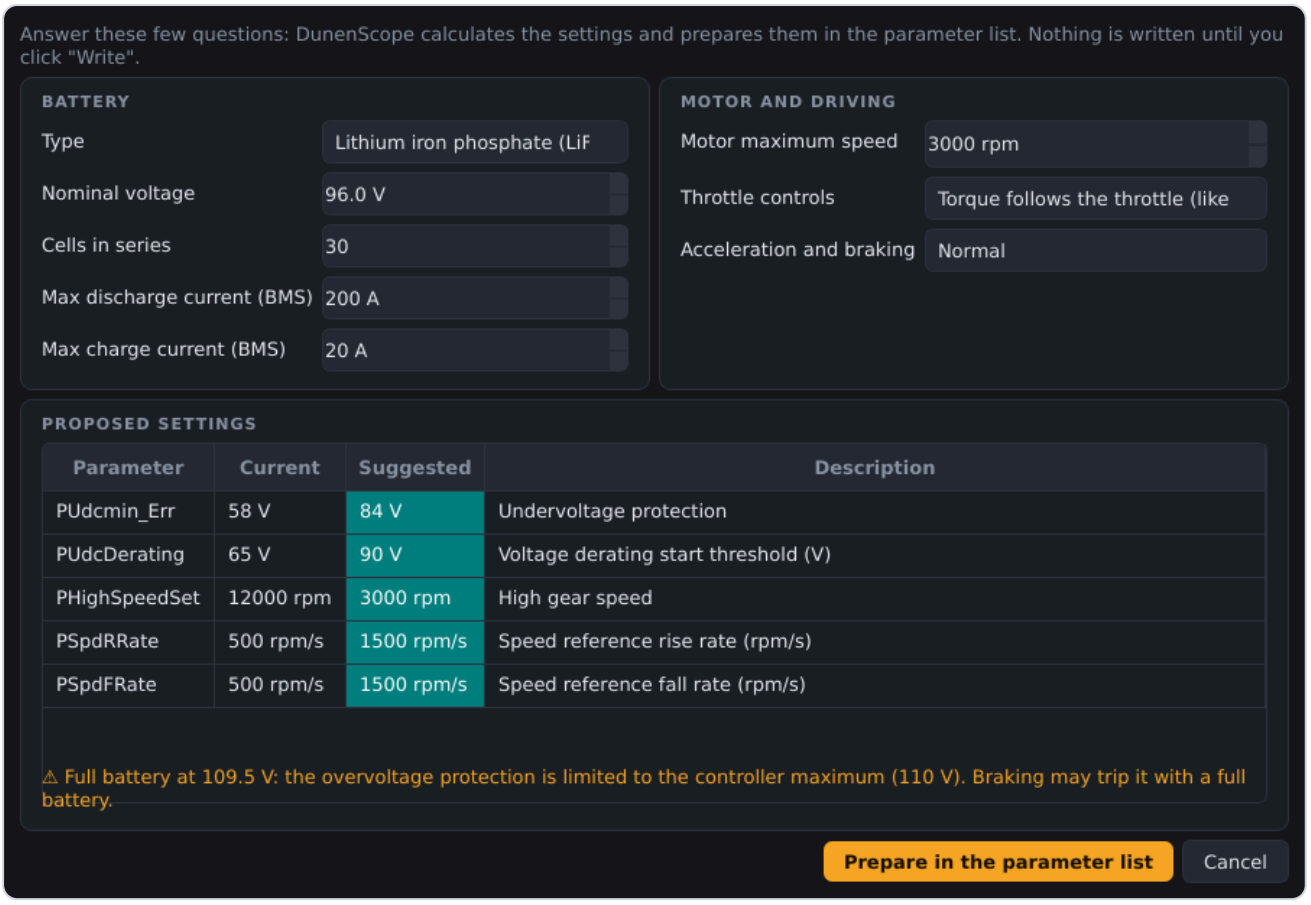
- **Read all parameters**: reads the full set; "Stop" interrupts the read.
- **Search and filters**: by name, description or row number; by group; settings and/or measurements; "Show hidden parameters" (greyed out); "Differences only".
- **Controller and Reference columns** side by side (the reference is the comparison file). The legend under the table explains the colours: blue ✎ to be written, orange different from the reference, red outside the file's Min / Max bounds (on the screenshot, 12000 rpm for a maximum of 10000), grey hidden in DunenConfiger. At the Simple level, the Min, Max and Type columns are hidden (the range is in the help panel).
- **Files menu**: choose the parameter file, compare with a file, export to xlsx (re-importable into DunenConfiger) or CSV.
- **Double-click the controller value**: change it directly. If needed, DunenScope offers to allow changes (with confirmation) and, for a sensitive setting shown at the Simple level, to switch to the Workshop level. Double-click elsewhere on the row: adds the variable to the custom plot on the Real time page.
- **Level**: *Simple* (default) shows only the twenty or so settings that are really changed; settings that affect the basic behaviour of the motor are only displayed there. *Workshop* shows every parameter. *Manufacturer* (special manufacturer key) also unlocks the parameters reserved for the manufacturer.
- **Help**: clicking a parameter shows, in the right-hand panel, its explanation, our advice, its current value, the file value, the factory value, the allowed range and the precautions.



Help

Easy setup

The "Battery assistant..." button (Controller and battery step of the Setup wizard, and Settings → Battery) asks a few questions: battery type and voltage (and number of cells), maximum discharge and charge currents allowed by the BMS, maximum speed, torque or speed control, soft, normal or sharp acceleration. DunenScope calculates the matching settings (undervoltage cut-off, power derating, overvoltage protection, current limits, overspeed, ramps, mode) and shows "current → proposed", with warnings when a value exceeds what the controller supports.



Easy setup

"Prepare in the parameter list" places the values in the list (rows marked ✎): nothing is written before "Write".

Changes allowed (expert)

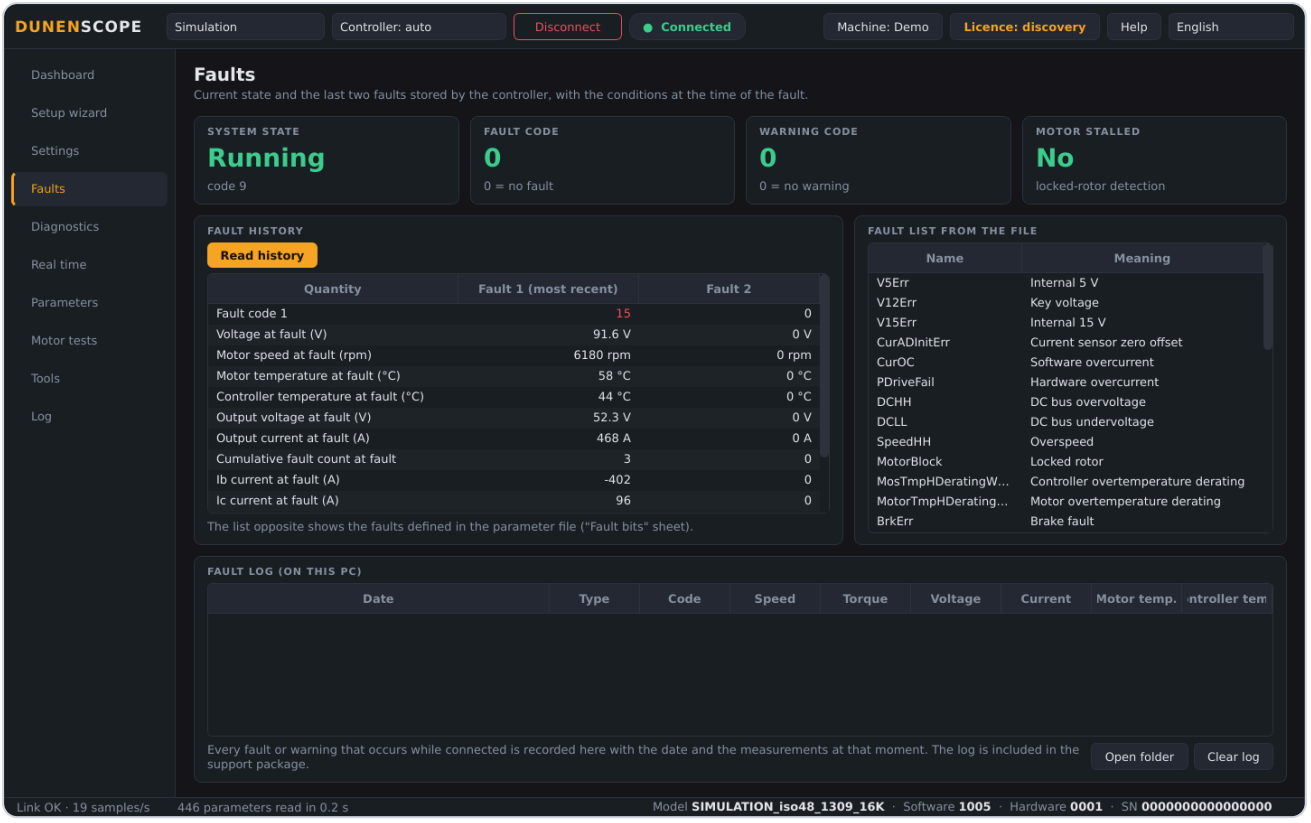
Tick "Changes allowed (expert)" and confirm. Values in the Controller column become editable. Each pending change is marked ✎ and counted on the "Write N change(s)" button; "Discard changes" drops them.

1. A value outside the file's bounds is rejected as soon as it is typed.
2. On "Write", DunenScope checks the consistency of the settings as they will be after writing: maximum speed above the overspeed threshold, throttle rest ≥ full, inconsistent voltage thresholds or thresholds too close to the current battery voltage, zero current or torque, zero speed gain in speed mode, derating temperatures too high... Serious problems (⛔) block the write at Simple level; at Workshop level, "Write anyway" must be confirmed. Points to check (⚠) are recalled in the confirmation.
3. DunenScope lists the changes and asks for confirmation, with a stronger warning for sensitive parameters (motor direction, pole pairs, encoder zero, protection thresholds...).
4. The motor must be stopped; otherwise the write is cancelled.
5. A full parameter backup is made just before. If it fails, nothing is written.
6. As with DunenConfiger, the whole settings table is rewritten, then the controller saves it in its permanent memory for 20 seconds: do not switch the key off before the final message. All settings are then read back for verification, and the log records each value as "written and verified".

Some parameters can never be changed: PBootNodeID and PBootCANBaudRat (CAN address and bit rate), PDefaultFn (restore defaults). Measurements (row 258 onwards) and factory calibration cannot be changed.

11. Faults

The Faults page shows the current state and the last two faults stored by the controller.



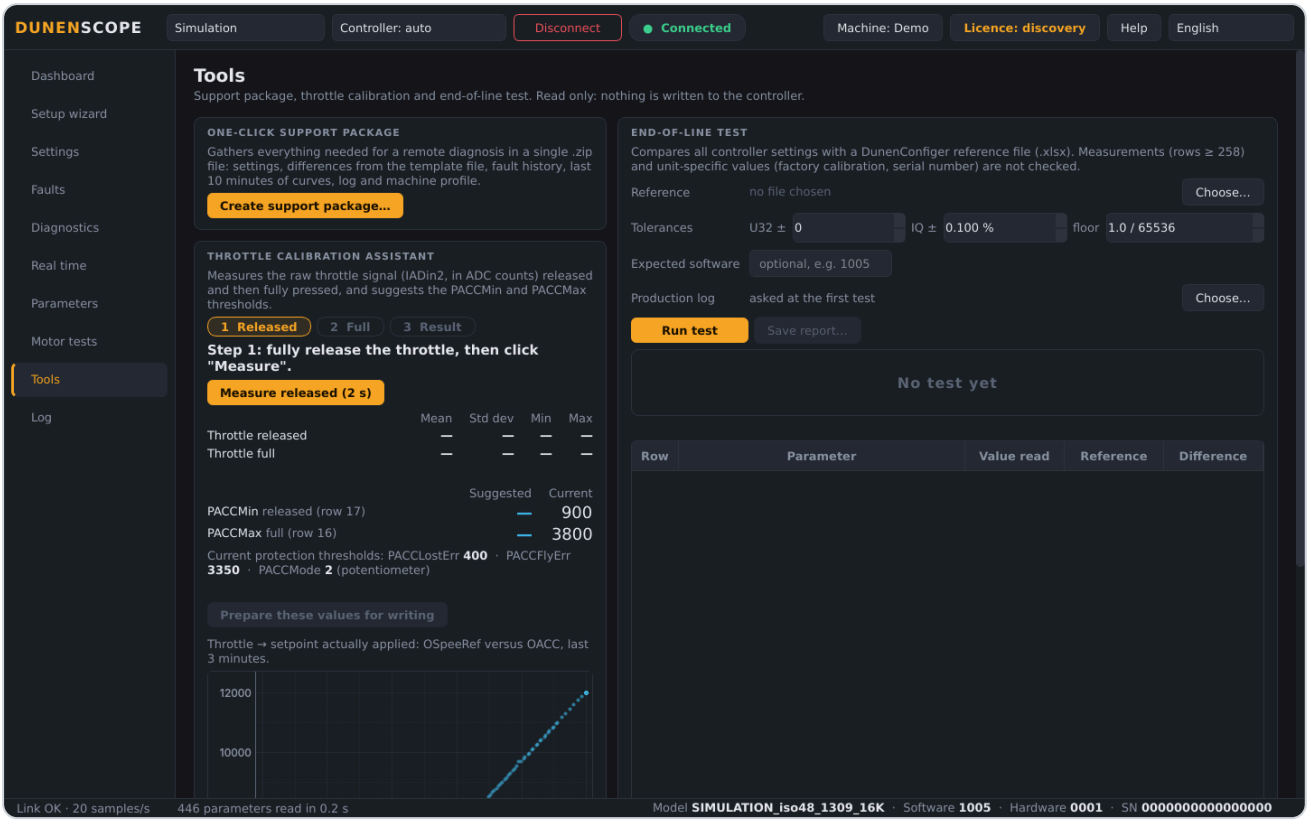
Faults

- **Top row**: system state, fault code, warning code, motor stall detection.
- **Fault history**: "Read history" shows, for the most recent fault and the one before, the code and the conditions at the time of the fault: voltage, speed, temperatures, output voltage and current, Ib and Ic currents, 5 V, 12 V and key voltages, cumulative fault count.
- **Fault list from the file**: the faults defined in the parameter file ("Fault bits" sheet) and their meaning.
- **Fault log (on this PC)**: the controller only keeps its last two faults, without a date. DunenScope therefore records every fault or warning that appears while connected, with the date, the controller and the measurements at that moment (speed, torque, voltage, current, temperatures). The log (Documents\DunenScope\journal_defauts.csv) is included in the support package; "Clear log" resets it.

For a guided analysis of the last fault, use the Diagnostics page.

12. Tools

The Tools page groups the support package, throttle calibration and the end-of-line test. It only reads from the controller.

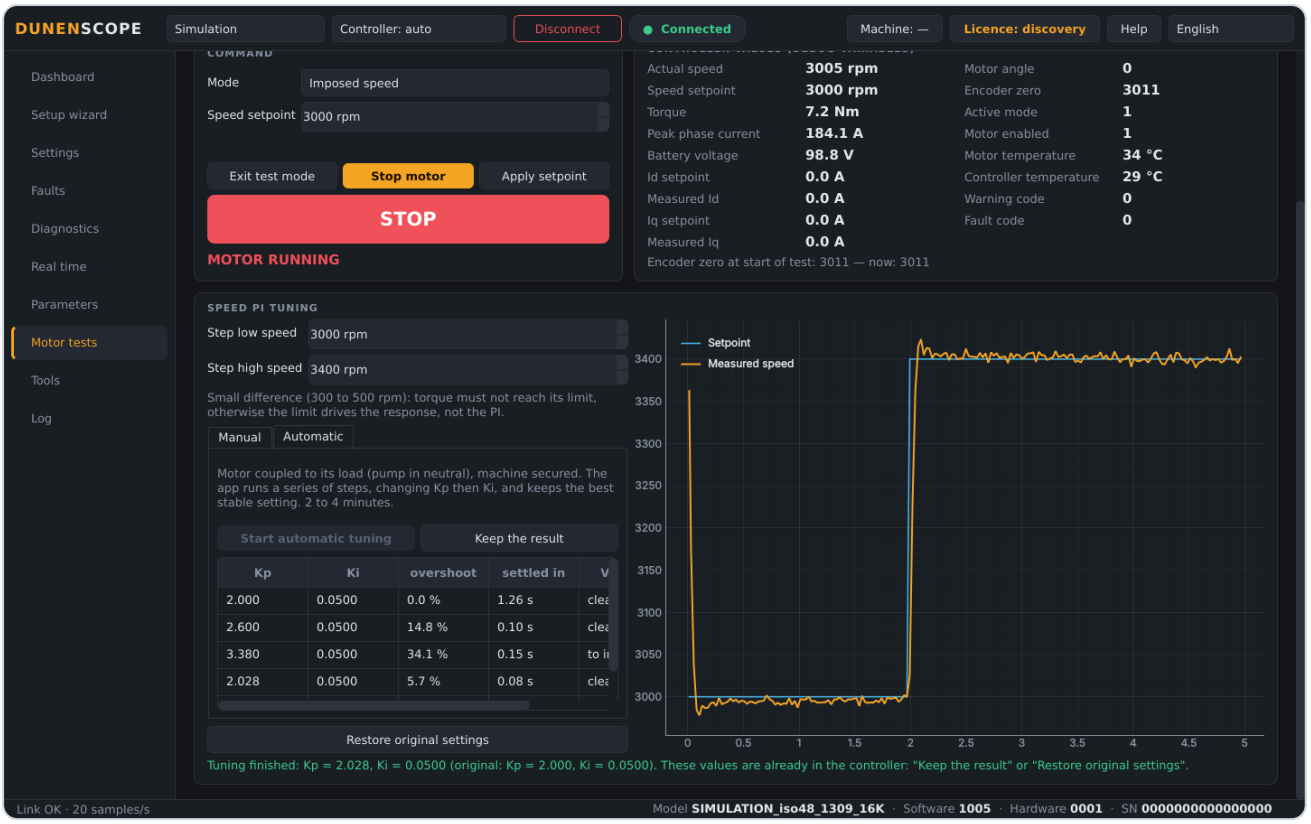


Tools

- **One-click support package**: "Create support package..." collects in a single .zip file the controller and machine information, all parameters, differences from the model file, fault history, plots of the last 10 minutes and the log. The customer sends it as is to support.
- **Throttle calibration assistant**: in three steps (Released, Full, Result), it measures the raw pedal signal (IADin2) released then fully pressed and proposes PACCMin and PACCMax, next to the current values and the protection thresholds. "Prepare these values for writing" places them on the Parameters page, where they are written in expert mode. A plot shows the speed setpoint actually applied as a function of the pedal over the last 3 minutes.
- **End-of-line test**: compares every controller setting with a DunenConfiger reference file (.xlsx), with tolerances (U32, IQ in %, floor) and the expected software version. Measurements (row 258 onwards) and unit-specific values (factory calibration, serial number) are not checked. "Run test" shows the differences row by row; "Save report..." produces an HTML report (with a CSV version). Each test can be added to a production log.

13. Motor tests and PI tuning

The Motor tests page drives the motor directly from the PC, like the "AC motor debug" page of DunenConfiger. It requires expert mode, enabled on the Parameters page. The screenshot shows a completed automatic PI tuning.



Motor tests

Before starting: motor decoupled or machine made safe (wheels lifted, pump in neutral, nobody nearby), then tick "The machine is safe and the motor can turn freely".

- **Modes**: simulated throttle, imposed speed, imposed torque, encoder angle auto-learning, encoder adjustment (only when instructed by Dunen), current loop (Id / Iq).
- **Command**: "Enter test mode", "Start motor", "Apply setpoint". The **STOP** button and the **Esc** key stop the motor immediately.
- **Controller values** (debug variables), read live: actual speed and setpoint, torque, peak phase current, battery voltage, Id and Iq (setpoint and measured), motor angle, encoder zero, temperatures, warning and fault codes. The encoder zero at the start of the test stays displayed to check the auto-learning.

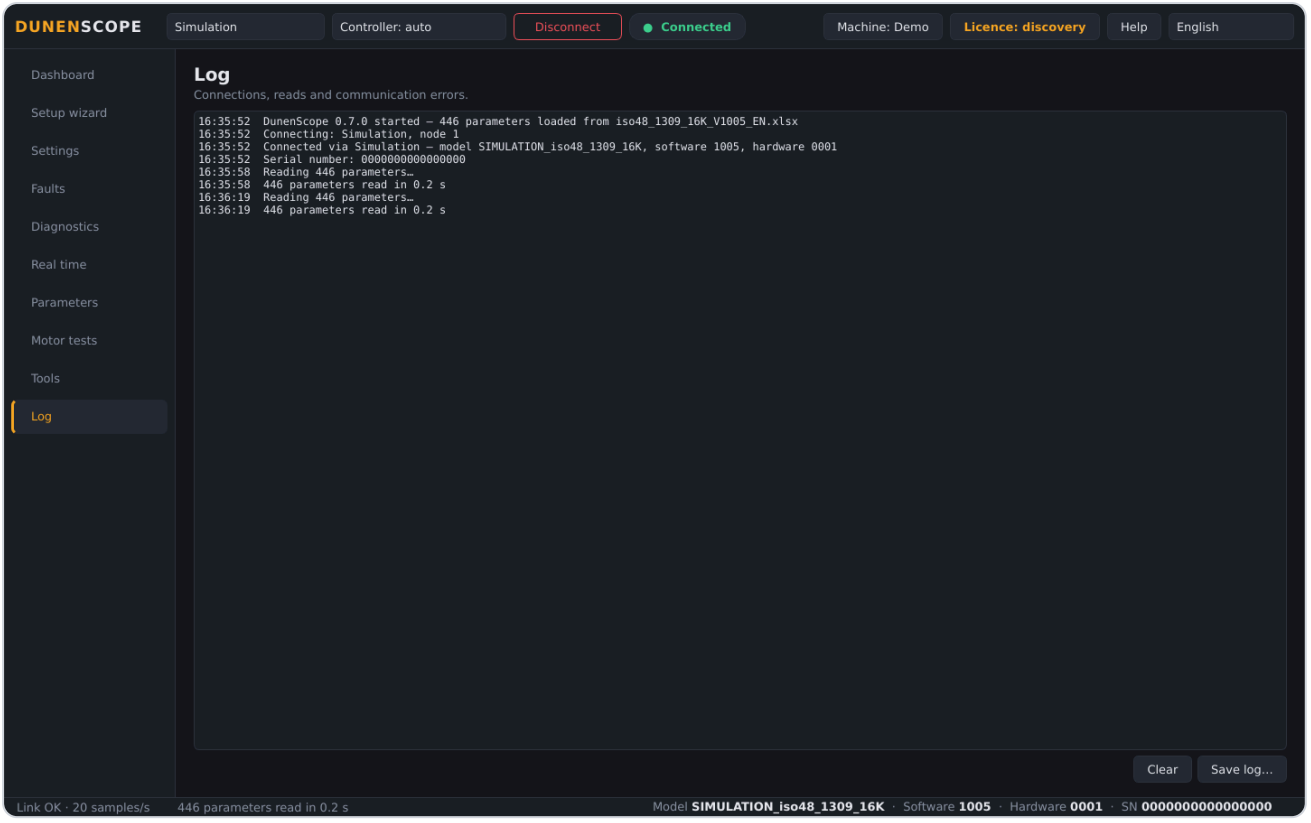
Speed PI tuning

Tuning uses a speed step, for example 3000 → 3400 rpm. A small difference (300 to 500 rpm) keeps the torque below its limit: otherwise the limit drives the response, not the PI.

- **Manual**: enter Kp and Ki, write them, run a step and read the response and recommendations.
- **Automatic**: motor coupled to its load (pump in neutral), machine secured. DunenScope runs a series of steps, changing Kp then Ki, and keeps the best stable setting, in 2 to 4 minutes. Each trial is listed with Kp, Ki, overshoot, settling time and verdict.
- The plot overlays the setpoint and the measured speed. On the screenshot: Kp = 2.028, Ki = 0.0500 (original: Kp = 2.000, Ki = 0.0500). These values are already in the controller; choose "Keep the result" or "Restore original settings".

14. Log

The Log timestamps the session's events: start-up and loaded parameter file, connection (adapter, node, model, versions, serial number), reads, backups, verified writes, motor tests and communication errors.

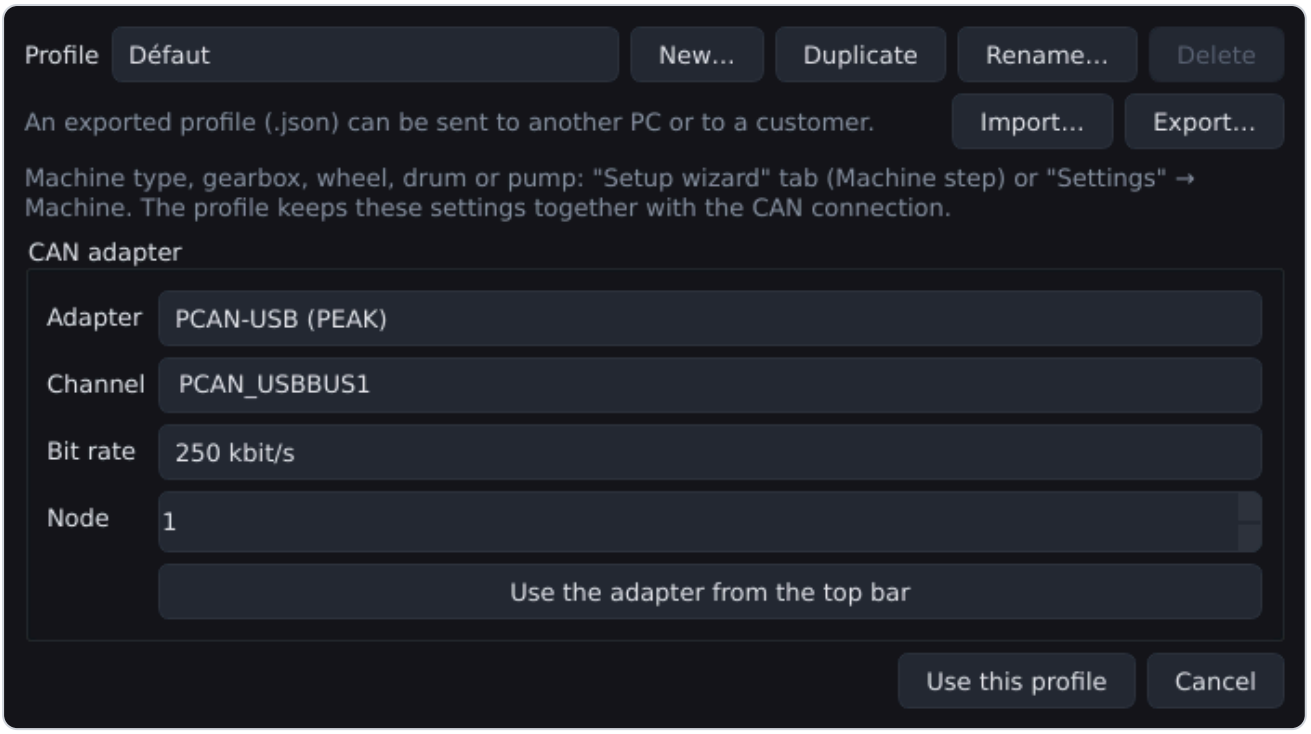


Log

"Save log..." saves it to a file to attach to a support request. "Clear" empties it.

15. Saved machines

The saved machines window opens from the "Machine: ..." button in the top bar. Each machine keeps its name, its description (set in Setup wizard → Machine or Settings → Machine) and the CAN adapter configuration.



Saved machines

- **Machine**: described in the Setup wizard (Machine step) or in Settings → Machine: machine type (generic, vehicle, hydraulic pump, winch or hoist, conveyor, fan or compressor), reduction ratio (1 = direct drive), wheel or drum diameter, pump displacement. These values are not written to the controller: they show speeds in the machine's units.
- **CAN adapter**: adapter, channel, bit rate, node ("Automatic" by default). "Use the adapter from the top bar" copies the current configuration.
- **Management**: New, Duplicate, Rename, Delete. "Export..." produces a .json file to send to another PC or to a customer, who opens it with "Import...".

16. Safety

By default, DunenScope only reads from the controller. Writing parameters and running motor tests require the full version (section 3) and changes to be allowed, with confirmation (banner of the Settings and Setup wizard pages, or the "Changes allowed (expert)" box on the Parameters page).

- **Automatic backup**: full backup before every write, in Documents\DunenScope\sauvegardes. If the backup fails, the write is cancelled.
- **Read-back** of every written value, compared with the requested value.
- **Dangerous parameters blocked**: CAN address and bit rate, restore defaults. Measurements and factory calibration are read-only. Out-of-range entries are rejected. Stronger confirmation for sensitive parameters.
- **Motor stopped**: required for any write.
- **Motor tests**: STOP or Esc = immediate stop. Automatic stop after 2 minutes of running, when leaving the page, or if the connection or expert mode is lost.
- **First use** of writing and motor tests: motor decoupled or machine made safe, hand on STOP.

After a write, check that the values are retained: switch the key off for 10 s, reconnect, read the parameters again and check that the written values are still there.

17. Technical specifications

Item	Value
Operating system	Windows 10 / 11, 64-bit
Protocol	Modbus RTU over CAN, same as DunenConfiger
CAN identifiers	0x602 (PC → controller), 0x6D2 (controller → PC), standard frames
Request size	24 registers maximum
CAN bit rate	250 kbit/s default; 125, 500, 1000 kbit/s
Node	1 by default (1 to 10)
Sampling rate	about 20 samples/s
CAN adapters	PEAK PCAN-USB, Kvaser, IXXAT, Vector, SLCAN (CANable…); Simulation mode without hardware
Controller data	446 parameters, 25 debug variables
Languages	French, English, Simplified Chinese, Spanish, German, Italian, Portuguese (Brazil), Russian, Japanese, Korean
Licence	Free discovery mode; full version with a key (lifetime or subscription, one PC per key); signed updates
Exports	DunenConfiger-compatible xlsx; CSV (parameters, measurements); support package .zip; end-of-line HTML report

18. Release status

Version 0.7.0: "Setup wizard" (motor, controller, command, machine, writing, encoder zero, direction of rotation, PI), "Settings" page by category in the machine's units, mouse-editable curves, winch, conveyor and fan machine types. Version 0.6.0: automatic connection, "Other controller" mode, easy setup, Simple / Workshop / Manufacturer levels, help on every parameter, consistency check before writing, fault log, Help button to write directly to the publisher, error reports. Version 0.4.2: online purchase open (lifetime, yearly, quarterly) and customer portal; updates free for everyone. Version 0.4.0: licensing (discovery mode, online purchase, publisher keys), signed automatic updates, interface and parameter files in 10 languages. Parameter reading and writing used on a real controller (electrified IMER Taurus 1300 mini dumper); validation of motor tests and automatic PI tuning in progress.